Alexandroff Spaces

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Definitions and Examples

Definition 1.

Let $(X, \mathcal{A})$ be a topological space. We say that it is an **Alexandroff Topology (AT)** if for $(O)_i \in I$ indexed family of $\mathcal{A}$-open sets we $\bigcap_{i \in I} O_i$ is also $\mathcal{A}$-open.

- We can also, by duality, define *Alexandroff Topology* to be a topology where $\bigcup_{i \in I} C_i$ is also $\mathcal{A}$-closed for $(C)_i \in I$ indexed family of $\mathcal{A}$-closed sets.

Examples

The following spaces are Alexandroff spaces:

- The *Sierpiński* space $\mathcal{J} = (\{0, 1\}, \mathcal{T})$ with $\mathcal{T} = \{\phi, \{0\}, \{0, 1\}\}$
- A *Colpen topology* which is a topology in which every open set is also closed, and vice versa.
- Any topological space equipped with the discrete topology.

Minimal Open Neighborhood

Definition 2.

Let $(X, \mathcal{T})$ be a topological space, and let $x \in X$. We say that x has a **minimal open neighborhood**, denoted by $V(x)$, if the intersection of all open sets containing x is itself an open set. That is,

$$V(x) = \bigcap_{\substack{O \in \mathcal{T} \\ x \in O}} O,$$

and $V(x)$ is open.

Theorem 1.

Let $(X, \mathcal{T})$ be a topological space, then $(X, \mathcal{T})$ is *Alexandroff* space if and only if each point in X has a minimal open neighborhood.

Minimal Open Neighborhood

Proposition 1.

Let $(X, \mathcal{T})$ be a T_1 topological space. Then the following statements are equivalent:

- X is an *Alexandroff* space.
- $\mathcal{T}$ is the discrete topology.

Proof. Assume X is an Alexandroff space.
Since X is T_1, for each $x \in X$, the singleton $\{x\}$ is closed.
So $X \setminus \{x\}$ is open, and the intersection

$$\bigcap_{y \neq x} (X \setminus \{y\}) = \{x\}$$

is also open. Thus, every singleton $\{x\}$ is open.
Therefore, every subset of X is open, i.e., $\mathcal{T}$ is the discrete topology.

Preorders Induced by a Topology

Definition 3.

Let $\leq$ be a binary relation on X, then

- $i)$ $\leq$ is called **preorder** on X if $\leq$ is both reflexive and transitive.
- $ii)$ $\leq$ is called **a partial order** on X if $\leq$ is preorder and for $x, y \in X$ such that $x \leq y$ and $y \leq x$, we have $x = y$.

Proposition 2.

Let $(X, \mathcal{T})$ be topological space. Define a relation $\leq$ on X by

$$x \leq y \text{ if and only if } x \in \overline{\{y\}}$$

then $\leq$ is a preorder on X called the **specialization preorder** induced by the topology.

Preorders Induced by a Topology

Proposition 3.

Let $(X, \mathcal{A})$ be an Alexandroff space, then X is T_0 if and only if $\leq$ is a partinal order in X.

proof. $(\Rightarrow)$ We show that $\leq$ is: **Antisymmetric:** Suppose $x \leq y$ and $y \leq x$. Then $x \in \overline{\{y\}}$ and $y \in \overline{\{x\}}$, which implies $\overline{\{x\}} = \overline{\{y\}}$. Since X is T_0, no two distinct points have the same closure, so $x = y$. Therefore, $\leq$ is a partial order.
$(\Leftarrow)$ Suppose $x \neq y \in X$. Since $\leq$ is antisymmetric, at least one of $x \nleq y$ or $y \nleq x$ holds.
Assume without loss of generality that $x \nleq y$. Then $x \notin \overline{\{y\}}$, which implies that there exists an open set $U \ni x$ such that $y \notin U$. therefore X is T_0. □

Alexandroff Topology Induced by a Preorder

Let $\leq$ be preorder on a set X, and let

$$\mathcal{B}(\leq) := \{(\downarrow x] : \quad x \in X\}$$

, with

$$(\downarrow x] := \{y \in X : \quad y \leq x\}$$

Lemma 1.

The set $\mathcal{B}(\leq)$ is a basis of a topology, denoted by $\mathcal{A}(\leq)$.

Lemma 2.

Let $U \subseteq X$. Then, U is a $\mathcal{A}(\leq)$- open if and only if $\forall x \in U$, $(\downarrow x] \subseteq U$

Theorem 2.

$\mathcal{A}(\leq)$ is an Alexandroff topology.

Proof. By **Lemma 1**, $\mathcal{A}(\leq)$ is a topology on X generated by the basis $\mathcal{B}(\leq)$. Let $\{U_i\}_{i\in I}$ be a collection of open sets in $\mathcal{A}(\leq)$, and let U $:= \bigcap_{i\in I} U_i$. Let $x \in U$. Then $x \in U_i$ for all $i \in I$. Since each U_i is open, Lemma 2 implies $(\downarrow x] \subseteq U_i$ for all i. Hence,

$$(\downarrow x] \subseteq \bigcap_{i\in I} U_i = U.$$

Therefore, by Lemma 2 again, U is open in $\mathcal{A}(\leq)$. Hence,A($\leq$) is an Alexandroff topology. □

Continuous Functions between Alexandroff Spaces

Theorem 3.

Let $f : (X, \mathcal{A}) \longrightarrow (Y, \mathcal{A}')$ be a function between topological spaces. Then the following statements are equivalent:

- f is continuous.
- For all $x, y \in X$, if $x \leq_{\mathcal{A}} y$, then $f(x) \leq_{\mathcal{A}'} f(y)$.

Connectedness of Alexandroff Spaces

Theorem 4.

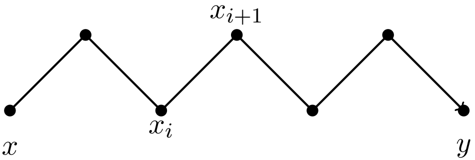
If $(X, \mathcal{A})$ is Alexandroff, then X is locally connected.

Theorem 5.

Let $(X, \mathcal{A})$ be Alexandroff space, then X is connected if and only if

$$\forall x, y \in X, \quad \exists x_o, x_1, \cdots, x_n$$

with $x_o = x, x_n = y$ such that $x_i \leq x_{i+1}$ or $x_{i+1} \leq x_i$.



Connectedness of Alexandroff Spaces

Theorem 6.

Let $(X, \mathcal{A})$ be an Alexandroff space. Then the following statements are equivalent:

- X is connected.
- X is path-connected.

References

Rose, D., Scible, G., Walsh, D. (2012). *Alexandroff Spaces*. Journal of Advanced Studies in Topology, 3(1), 32. https://doi.org/10.20454/jast.2012.233. Courier Corporation.